

Sheet 1: Finite Difference Method

1- Prove that the *FD* numerical derivatives are equivalent to fit a polynomial to data and then taking the derivatives (use example of 3 points data).

2- Explain the steps to get a numerical FD model from a physical model. Explain sources of error in each step.

3-Discuss the different sources of error in the computational methods. Then use the FD central formula for $\frac{\partial \phi}{\partial x}$ to show the effect of decreasing Δx on the truncation error.

4- The Finite Difference method is used to solve the damped wave equation:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} - a \frac{\partial u}{\partial t}$$

Where: $0 < x < 1$ and $t > 0$, $c=1$, $a=2$

Initial displacement $u(x,0) = \sin \pi x$, initial velocity $v_0 = \frac{\partial u}{\partial t} = 10 \text{ cm/s}$,

Boundary conditions: $u(0,t) = 0$ and $u(1,t) = 0$

Let $\Delta x = 0.2 \text{ cm}$, $\Delta t = 0.05 \text{ sec}$

5- Find the FD formula for the diffusion equation, $k \frac{\partial \phi}{\partial t} = \frac{\partial^2 \phi}{\partial x^2}$

Using the forward difference formula for $\frac{\partial \phi}{\partial t}$ and the central difference formula for $\frac{\partial^2 \phi}{\partial x^2}$.

Then prove that the obtained formula is valid only for $0 < \frac{\Delta t}{k(\Delta x)^2} \leq 0.5$.

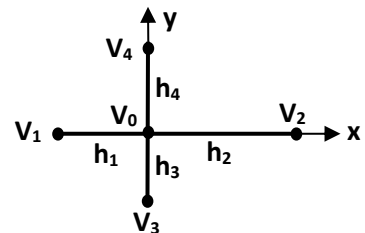
6- The one-dimensional wave equation is given by :

$$\frac{\partial^2 \phi}{\partial x^2} = \frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2}$$

Get an equivalent finite difference formula in terms of the aspect ratio $r = \left[\frac{c \Delta t}{\Delta x} \right]^2$.

Draw the computational molecule in this case and in case $r = 1$.

7- Find the finite difference approximation of Laplace's equation $\nabla^2 V = 0$ for the computational molecule shown in the figure.

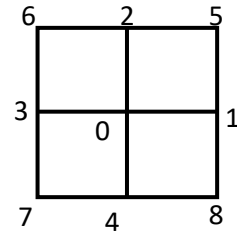


8- Prove that the FD₂₀ formula, given by,

$$4 \sum_{r=1}^4 \phi_r + \sum_{r=5}^8 \phi_r - 20\phi_0 = \left(-6k_c^2 h^2 + \frac{1}{2}k_c^4 h^4\right)\phi_0 \text{ is an}$$

FD approximation of Helmholtz equation, $\nabla^2 \phi + k_c^2 \phi = 0$,

for the computational molecule shown in the figure.



9- Prove that the formula,

$$\frac{\phi_{i+2} - 2\phi_{i+1} + 2\phi_{i-1} - \phi_{i-2}}{2\Delta x^3}$$

is a FD approximation for $\frac{d^3 \phi}{dx^3}$. Determine the order of the error.

10- Explain how to solve numerically a linear system of equations $[A][X] = [b]$, using the SOR method.

11- Consider an oscillator of mass $m = 1$, which obeys the equation of motion,

$$\frac{d^2 \theta}{dt^2} = -\sin \theta$$

If it has an initial displacement $\theta_0 = \pi/6$ and an initial velocity $v_0 = 10 \text{ cm/s}$, at $t = 0$. Calculate $\theta(t)$ numerically using a suitable FD formula in the time interval $0 < t < 1$ sec, at steps $\Delta t = 0.2$ sec.

12--Prove that the formula:

$$\frac{-\phi_{i+2} + 16\phi_{i+1} - 30\phi_i + 16\phi_{i-1} - \phi_{i-2}}{12(\Delta x)^2}$$

Is a fourth order FD approximation for $\frac{\partial^2 \phi}{\partial x^2}$,

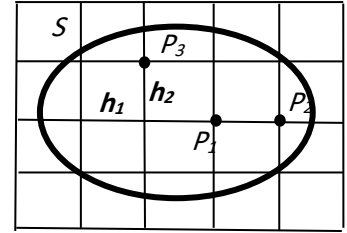
Use this formula to get a fourth order FD approximation for Laplace's equation in two-dimensional case. Let $\Delta x = \Delta y = h$. Draw the corresponding computational molecule.

13- According to the law of radioactive decay, the rate of decrease of the number of nuclei $N(t)$ of a particular radioactive isotope is given by:

$$\frac{dN(t)}{dt} = -\frac{N(t)}{\tau}$$

Where τ is the average lifetime of the isotope. Derive a suitable FD central formula to represent this formula, then starting with $N(0) = 104$ nuclei at $t = 0$ for an isotope with $\tau = 10$ seconds, calculate $N(t)$ in 10 time iterations $\Delta t = 1.0$ seconds.

14- The Poisson's equation $\nabla^2 \phi = g(x, y)$ is required to be solved numerically using FD method in a two dimensional region R surrounded by a contour S as shown in the figure. The region is divided into subdivisions so that $\Delta x = h_1$ and $\Delta y = h_2$ as shown.



Use central formula to represent the derivatives.

Write down the FD equations for the following points,

- An internal point as P_1 in R .
- A point P_2 near S , if points on S satisfy Dirichlet B.C. given by $\phi = C_1$.
- A point P_3 near S , if points on S satisfy Neumann B.C. given by $\frac{d\phi}{dy} = C_2$

15- Repeat problem (15) for Helmholtz equation $\nabla^2 \phi + k^2 \phi = 0$

16- Consider a particle of mass m which is moving along x axis and experiences a restoring force $-Cx$. If it is driven by an external harmonic force given by $F = F_0 \cos(\omega t)$, the equation of motion in this case has the form,

$$m \frac{d^2 x}{dt^2} + Cx = F_0 \cos(\omega t)$$

if $m=1$, $C=1$, $F_0=1$ and $\omega=2\pi$, use a suitable FD formula the approximate position of the particle $x(t)$ in the time interval $t=[0,1]$, $\Delta t = 0.1$, $x(0) = 0$, $\left. \frac{dx}{dt} \right|_{t=0} = 2.0 \text{ cm/s}$.

17- A rod of length $L = 2\text{m}$ with its ends in contacts with cold water at 10°C . If the rod initially at 100°C .

i-Use central difference approximation for $\frac{\partial^2 \phi}{\partial x^2}$ and forward difference approximation for $\frac{\partial \phi}{\partial t}$

to get the required FD approximation in terms of the aspect ratio, $r = \frac{\Delta t}{k(\Delta x)^2}$.

(Diffusion equation:

$$\frac{\partial^2 \Phi}{\partial x^2} = k \frac{\partial \Phi}{\partial t})$$

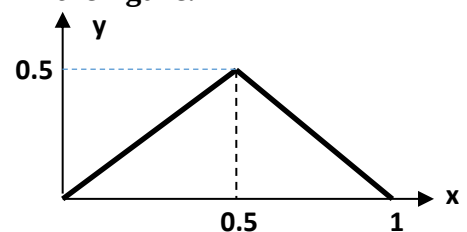
ii-Prove that this formula is stable only for $0 < r \leq 0.5$

iii-Use $\Delta x = 0.2 \text{ m}$, $\Delta t = 0.02 \text{ sec}$ and $k=1$ to get the temperature distribution in the rod at seven successive time iterations.

18- A vibrating string of unit length is fixed at both ends as shown in the figure.

If the initial displacement is given by:

$$y(x, 0) = x \quad \text{for } 0 \leq x < 0.5$$



$$y(x, 0) = 1 - x \quad \text{for} \quad 0.5 < x \leq 1.0$$

and the initial velocity, $\frac{\partial y}{\partial t}(x, 0) = 0$

Use $c=1$, $\Delta x = 0.1$ to get the displacement at seven time successive iterations in case of aspect ratio $r = 1$.

19- The equation of a critically damped harmonic oscillator is given by

$$m \frac{d^2 y}{dt^2} + c \frac{dy}{dt} + ky = 0 \quad \text{Where } c^2 = 4mk$$

If $c = 2$, $m = 1$, use FD method to get the values of ten time iterations ($\Delta t = 0.2$ sec) for the following initial conditions: $y(t=0) = 1.0$ m, $v_0 = \frac{dy}{dt}(t=0) = 0.5$ m/s.

20- A vibrating string of length

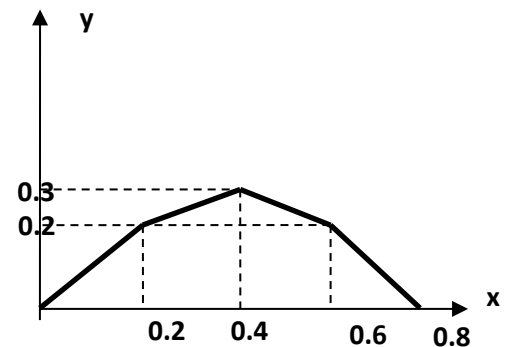
0.8 cm is fixed at both ends and stretched so that the initial displacement is as shown in the figure.

If the initial velocity = $\frac{\partial y}{\partial t}(x, 0) = 0$,

Derive a suitable FD formula to get the displacement

At $x = n \Delta x$, ($n=1,2,\dots,7$), $\Delta x = 0.1$ cm, for seven time successive iterations $\Delta t = 0.1$ sec. Use aspect ratio $r = 1$.

(Wave equation: $\frac{\partial^2 \phi}{\partial x^2} = \frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2}$, let $c = 1$ cm/s)



21- A pendulum of length l and a charged sphere of mass m swings in the presence of an oscillating electric field of angular frequency ω_d . If the amplitude of the electric force is q_1 and the frictional force is proportional to the velocity with proportionality factor q_2 .

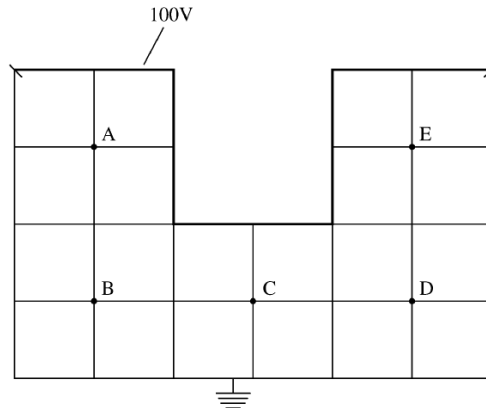
(i) Write down the equations of motion of the pendulum.

(ii) derive a finite difference algorithm to compute the swing angle θ and the angular frequency ω at discrete time intervals Δt with initial conditions θ_0 and ω_0 .

(iii) Using the values $l = 1.2$ m, $m = 50$ gm, $\omega_d = 1.2$ rad/s, $q_1 = 0.5$ N, $q_2 = 0.5$ N.s/m,

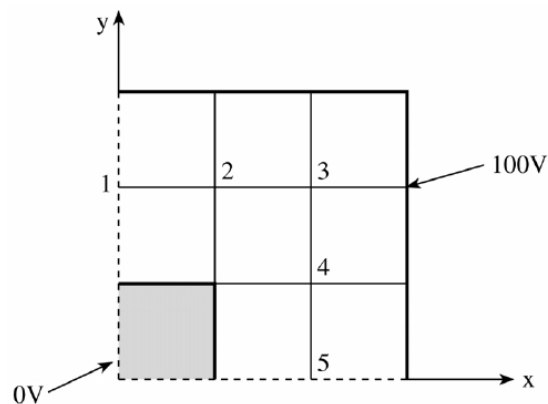
$\Delta t = 0.04$ s, $\theta_0 = 0.2$ rad and $\omega_0 = 0$, compute θ and ω at time $t = 0.12$ sec.

22- For a long hollow conductor with a uniform U-shape cross-section shown in the Fig. Derive a suitable FD formula to get the potential at points A, B, C, D, and E. ($\Delta x = \Delta y = 0.1$).



23- The potential system shown in the Fig. is a quarter section of a transmission line. Derive a suitable FD formula to get the potential at at nodes 1, 2, 3, 4, and 5. ($\Delta x = \Delta y = 0.1$).

Then find the electric field at point 2.



24- The wave equation in a stretched string is given by: $\frac{\partial^2 \phi}{\partial x^2} = \frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2}$, where $c = \sqrt{\frac{T}{\mu}}$, T is the tension, μ is the mass per unit length. In case of harmonic oscillations ($\frac{\partial}{\partial t} = j\omega$), Use a suitable FD formula to find the first five resonance frequencies. (use $L = 1.0$ m, $T = 2$ N).